Std. IX GEOGRAPHY

UNIT : 2

IN THE EXPANSIVE PLAIN

NORTH INDIAN PLAIN

- The vast physical division that lies to the south of the Northern Mountains and to the north of the Peninsular Plateau.
- It is also known as Indo -Gangetic- Brahmaputra Plain.

FORMATION OF PLAIN

- The fertile Indo- Gangetic-Brahmaputra plain was formed by the deposition of sediments carried by rivers.
- The sediments are deposited in a vast depression at the south of the Himalayas which was formed as a result of the formation of the Himalayas.
- The sediments, transported by rivers originating from both the Himalayas and Peninsular India, were deposited in this expansive depression, which led to the formation of the plain.
- The continuous depositional processes over millions of years have contributed to the evolution of this fertile plain.
- The average depth of alluvial deposits in this plain ranges from 1000 metres to 2000 metres.

FEATURES OF NORTH INDIAN PLAIN

- This Plain extending (Length) approximately over 3200 km.
- It is one of the largest alluvial plain in the world.
- It spreads (length) over around 2400 km in India.
- The plain widens from east to west.
- The width ranging from 150 km to 300 km.
- This plain is bordered by the Shiwalik ranges in the north and the Peninsular Plateau in the south.
- The plain covers an area of approximately 7 lakh sq.km.
- Features such as fertile soil, water supply, favourable climate and flat topography make this region suitable for agriculture.

REGIONAL DIVISIONS OF NORTH INDIAN PLAIN

1. Rajasthan Plains
2. Punjab-Haryana Plain
3. The Ganga Plain
4. Brahmaputra Plain

Rajasthan Plain

- It is the western most end of the North Indian Plain.
- The Rajasthan Plain is situated to the west of the Aravali Mountain range.
- Two-thirds of the Thar Desert is situated in Rajasthan.

- Remaining portion extends into the neighbouring states of Haryana, Punjab, and Gujarat.
- The Thar Desert is divided into two regions:
  1. Marusthali (the actual desert area)
  2. Bagar (semi desert region)
- Luni is the major river in this plain.
- The major salt lakes are Sambhar, Didwana and Sargol.

Punjab-Haryana Plain

- The Punjab-Haryana plain is situated to the east of the Rajasthan Plain.
- The eastern border of this plain is River Yamuna.
- In India, this plain extends over the states of Punjab, Haryana, and Himachal Pradesh.
- The plain covering an area of approximately 1.75 lakh sq.km
- The Punjab plain is formed by the deposition of sediments carried by rivers such as Satluj, Jhelum, Chenab, Ravi and Beas.
- Punjab is known as the Land of Five Rivers.
- The Punjab-Haryana Plain is divided into five major doabs.
- A doab is a land lying between two rivers that join together later.

1. Bist-Jalandhar Doab
2. Bari Doab
3. Rachna Doab
4. Chaj Doab
5. Sindh-Sagar Doab

The Ganga Plain

- Situated to the east of the Punjab - Haryana Plain.
- The Ganga Plain stretches from Bangladesh in the east to the Yamuna River in the west.
- Approximately 3.75 lakh sq.km area is covered.
- This plain covers the states of Uttarakhand, Uttar Pradesh, Haryana, Delhi, Jharkhand and West Bengal.
- This plain was formed through the depositional processes by River Ganga and its tributaries.
- The average elevation of the Ganga Plain is 200 metres above mean sea level.
- The Ganga plain divided into three regions:-
  1. Upper Ganga Plain
  2. Middle Ganga Plain
  3. Lower Ganga Plain

Brahmaputra Plain

- Constitutes the eastern most part of the North Indian Plain.
- The Brahmaputra Plain, known by various names such as Brahmaputra Valley, Assam Valley and Assam Plain.
- This plain Stretching from the eastern most edge of Assam to the west of Dubri, near the border of Bangladesh.
- Approximately 720 km in length, and width ranging from 60 to 70 km.
- The major portion of the Brahmaputra Plain is located in the state of Assam.

- Natural boundaries of Brahmaputra Plain:-

  North : Eastern Himalaya
  East : The Patkai-Naga Hills
  South : Garo-Khasi-Jaintia Hills and Mikir Hills
  West : The Lower Ganga Plain
- Approximately 56,275 sq.km area is covered.
- Is formed through the depositional processes carried out by the Brahmaputra River and its tributaries.
- Teesta, Manas, Lohit and Dibang are the major tributaries of the Brahmaputra River.
- This plain is rich in alluvial fans.
- When rivers enter a plain, their velocity decreases rapidly. The sediments carried by the rivers get deposited in the form of fans. Such depositional landform features are referred to as alluvial fans.
- When the flow is too low, the river flow in sinuous manner and then they form oxbow lakes.
- A river flowing in a sinuous manner is called river meandering.

The North Indian Plain can be divided into three zones from north to south based on its geomorphic features:-

1. Bhabar
2. Tarai
3. Alluvial plains

Bhabar

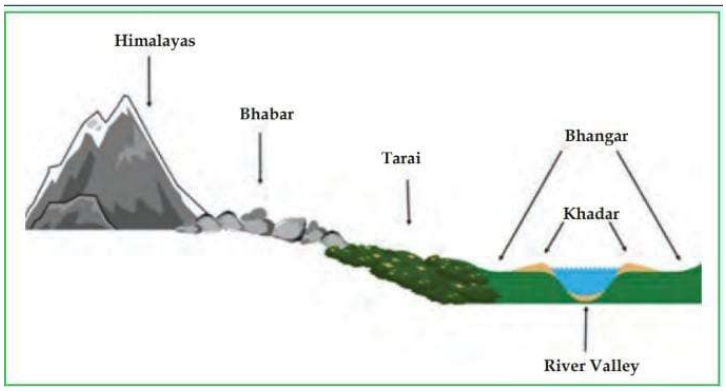
- Bhabar is situated parallel to the south of Shiwalik mountain range.
- Bhabar is a narrow belt, approximately 8 to 10 km wide.
- This zone of formed by the deposition of rocks and boulders brought by rivers flowing from the mountains.
- The rivers flowing through this region are not visible as they flow under the rocks and boulders.

Tarai

- The Tarai is a marshy and swampy tract, seen parallel to the Bhabar belt.
- Approximately 10 to 20 km wide.
- The rivers that disappear in the Bhabar region re- emerge in Tarai region.
- The Tarai has a luxurious growth of natural vegetation and serves as a habitat for varied wildlife.

Alluvial plain

- To the south of the Tarai, the belt consisting of older and newer alluvial deposits forms the Alluvial plain.
- The older alluvium deposits are called the "Bhangar" and the newer ones are called "Khadar".
- Riverine islands, sandbars and deltas are the depositional landforms seen here.
- Braided streams, meanders and ox-bow lakes are also prominent features of this area.



Climate of the North Indian Plain

- The Cold Weather Season
- The Hot Weather Season
- The Southwest Monsoon Season
- The Northeast Monsoon Season

The Cold Weather Season

- Usually, the cold weather season sets in by mid-November in North India.
- December and January are the coldest months in the North Indian Plain.
- Excessive cold is experienced during this period.

Why there is excessive cold weather experienced in the North Indian Plain?

- The North Indian Plain's distance from the ocean.
- Snowfall in the nearby Himalayan ranges.
- Cold winds from West Asia.
- The apparent movement of the sun from the northern hemisphere towards the southern hemisphere.
- Slight rainfall receives in North Indian Plain During cold weather season.

The Hot Weather Season

- The summer season in the North Indian Plain is experienced in the months of April, May and June.
- Summer is extremely severe in North Indian Plain.
- By the month of May, the temperature rises up to 48°C in the western part of the North Indian Plain.
- Hot, dry wind called 'Loo' blows from the desert region of Rajasthan to the Ganga Plain in the months of May and June increases the temperature considerably in the North Indian Plain.
- Dust storms are very common in the evenings in Punjab, Haryana, Rajasthan, UP.

The Southwest Monsoon Season

- By the month of March, the temperature increases in the Northern hemisphere.
- As a result, a low- pressure area is developed over the North Indian plain.
- This low-pressure area attracts the southwest monsoon winds to the Indian subcontinent.

- The southwest monsoon winds enter the Indian subcontinent as two branches:-

1. The Arabian branch
2. The Bay of Bengal branch

The Bay of Bengal branch of South West Monsoon Season

- The Bay of Bengal branch, entering the land through the Sundarbans delta region, gets bifurcated into two sub-branches.
- One branch moves eastward and enters the Brahmaputra Plain, caused widespread rains here.
- The Second branch, moving westward along the Ganga Plain, causes rainfall in West Bengal, Bihar, Uttar Pradesh, Delhi and proceeds further westward.
- In the Punjab Plain, this branch joins the Arabian branch which is moving parallel to the Aravali Mountains and then brings rain to the foothills of the Western Himalayas.
- The South- west monsoon season, experienced from june to September, is the main rainy season in the North Indian plain.

The North-east Monsoon Season

- By the end of September, as the sun apparently shifts towards the southern hemisphere, The low-pressure area that prevails over the North Indian Plain starts moving southwards.
- As a result intense high-pressure develops over the North Indian Plain and low pressure over the Indian Ocean.
- It causes wind to blow from North Indian Plain towards the Indian Ocean.
- As these moisture-less winds blow from the north-east, this season is called the northeast monsoon season.
- This season is named the retreating monsoon season.

October heat

- The North Indian Plain generally experiences a dry climate during North-east Monsoon season.
- Owing to the conditions of high temperature and atmospheric humidity, the weather becomes unbearable in North Indian plain. This phenomenon is commonly known as the 'October heat'

The Natural Vegetation of the North Indian Plain

- The prominent natural vegetation generally found in the Nirth Indian plains are:-
  1. The Tropical Deciduous Forests
  2. The Tropical Thorn Forests
  3. Swamp Forests

The Tropical Deciduous Forests

- The tropical deciduous forests are divided into two:-
  1. Dry deciduous forests.
  2. Moist deciduous forests.
- Teak, sal, shisham, mahua, amla and sandalwood are the main species of the tropical deciduous forests.

Dry deciduous forests

- Dry deciduous forests are found in regions where the annual rainfall ranges between 70 cm and 100 cm.
- In these forests, trees shed their leaves for approximately 6 to 8 weeks in the dry season.
- The dry deciduous forests are found in the plains of Uttar Pradesh and Bihar.

Moist deciduous forests

- The moist deciduous forests are found in region where the annual rainfall ranging from 100 cm to 200 cm.
- They are mainly seen along the strip of Shiwalik range, Odisha and West Bengal.

The Tropical Thorn Forests

- The natural vegetation in the western part of the North Indian Plain is sparse due to low rainfall and intensive cattle rearing.
- Tropical thorn forests are prevalent in the semi-arid areas of south-west Punjab, Haryana, Rajasthan, and Uttar Pradesh.
- These forests consist of various grasses and shrubs.
- Important species are babool, ber, wild date palm, khair, neem, khejri and palas.

The swamp forests

- The swamp forests are the natural vegetation found in the vast saline expanses of Rajasthan, freshwater lakes, the freshwater marshes of the Ganga Plain, the flood plains of the Brahmaputra, and in the deltaic region of Sundarbans.
- The marshy and expansive deltaic region of Ganga Plain in West Bengal is Sundarbans.
- The natural vegetation found luxuriously in this region is mangroves.
- This region serves as a natural habitat for the Royal Bengal Tiger.
- Sundri, a type of mangrove are unique feature of the sudharban delta.

The Major Soil Types of the North Indian Plain

1. The soil which is widespread in the North Indian Plain is alluvial soil.
   - There are two different types of alluvial soils developed in the Ganga Plain: Khadar and Bhangar.
   - Alluvial soil is well-suited for agriculture.

2. The red soil is found in the southern part of the Middle Ganga Plain.
   - It is the presence of iron in the soil that gives a red colour to it.

3. The soil found in the Sundarbans delta region is saline soil.
   - Salt content is high in this soil.
   - Saline soil consists of sand and loam.
   - Seawater intrusions into the deltas cause the formation of saline soils.

4. Along the coastal regions of West Bengal, peat soil is also found.

- Arid soil is the soil extensively found in the western parts of the North Indian Plain, including Rajasthan.
- It is generally sandy and saline in structure.
- As this soil lacks humus and moisture.
- Therefore Irrigation support is required for the plants to grow.

The Human Life

- Although this extensive plain accounts for less than one-fourth of the total area of our country, it is home to more than half of India's total population.
- Wheat, rice, jute, and sugarcane are widely cultivated here.
- Extensive cultivation supported by irrigation, has made this plain the granary of India.
- The entire plain, except the Thar Desert, has a well-developed network of roads and railways.
- Kharif, Rabi, and Zaid are the three different cropping seasons in the North Indian Plain.
- Kharif is associated with the southwest monsoon season.
- Rabi begins with the onset of the cold season.
- Zaid, a short duration cropping season, starts after the harvesting of Rabi crops.

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